4-5-23

Wednesday, April 5, 2023 10:28 AM

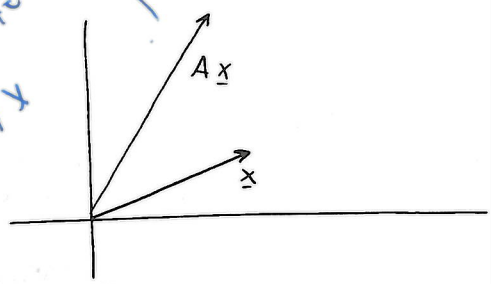
I)

eigenvalues

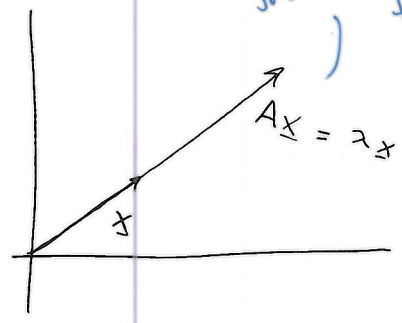
eigenvectors

$$\boxed{A\underline{x} = \lambda \underline{x}}$$

Typically the matrix will rotate or modify x



Depiction that eigenvectors are just scaled in the same direction



[Fact: If the eigenvalues of a $m \times m$ matrix A are distinct, then eigenvectors are linearly independent

Suppose A has distinct eigenvalues

$$\lambda_1, \lambda_2, \ldots, \lambda_m$$

$$A\underline{x}_1 = \lambda_1 \underline{x}_1 \qquad A\underline{x}_2 = \lambda_2 \underline{x}_2 \cdots \qquad A\underline{x}_m = \lambda_m \underline{x}_m$$

$$[A\underline{x}_1 \quad A\underline{x}_2 \quad \cdots \quad A\underline{x}_m] = [\lambda_1 \underline{x}_1 \quad \lambda_2 \underline{x}_2 \quad \cdots \quad \lambda_m \underline{x}_m]$$

$$A\underbrace{[\underline{x}_1 \ \underline{x}_2 \ \cdots \ \underline{x}_m]}_{S} = \underbrace{[\underline{x}_1 \ \underline{x}_2 \ \cdots \ \underline{x}_m]}_{S} \underbrace{\begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_m \end{bmatrix}}_{\Omega}$$

(Similarity matrix)

A is similar to Ω

(Apparently this was in linear algebra)

$$AS = S\Omega$$

$$\boxed{A = S\Omega S^{-1}} \qquad S^{-1}AS = \Omega$$

$$A^2 = S\Omega \cancel{S^{-1}} \cancel{S} \Omega S^{-1}$$

$$= S\Omega^2 S^{-1}$$

$$\vdots$$

$$\Omega^n \quad \begin{bmatrix} \lambda & & \end{bmatrix}^n$$

Convient way to compute power of matrices

$$A^n = S \Lambda^n S^{-1}$$

$$\Lambda^n = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_m \end{bmatrix}^n = \begin{bmatrix} \lambda_1^n & & \\ & \lambda_2^n & \\ & & \ddots \\ & & & \lambda_m^n \end{bmatrix}$$

Substitute

A) Unique Eigenvalues

Recall: $e^x = \sum_{i=0}^{\infty} \frac{x^i}{i!}$ ← Taylor series of e^x

Matrix exponential → $e^A = \sum_{i=0}^{\infty} \frac{A^i}{i!}$ → In MATLAB expm

$$= \sum_{i=0}^{\infty} \frac{S \Lambda^i S^{-1}}{i!}$$

$$= S \left(\sum_{i=0}^{\infty} \frac{\Lambda^i}{i!} \right) S^{-1}$$

$$e^A = S \begin{bmatrix} e^{\lambda_1} & & & \\ & e^{\lambda_2} & & \\ & & \ddots & \\ & & & e^{\lambda_m} \end{bmatrix} S^{-1}$$

If we have a polynomial, we can do a similar thing

$$p(t) = a_0 + a_1 t + \dots + a_s t^s$$

$$p(A) = a_0 I + a_1 A + a_2 A^2 + \dots a_s A^s$$

$$= a_0 S S^{-1} + a_1 S \Lambda S^{-1} + a_2 S \Lambda^2 S^{-1} + \dots + a_s S \Lambda^s S^{-1}$$

$$= S \left[a_0 I + a_1 \Lambda + \dots + a_s \Lambda^s \right] S^{-1} = S\, p(\Lambda) S^{-1}$$

$$= S \begin{bmatrix} p(\lambda_1) & & & \\ & p(\lambda_2) & & \\ & & \ddots & \\ & & & p(\lambda_m) \end{bmatrix} S^{-1}$$

B) <u>Repeated</u> <u>Eigenvalues</u>

$$I \underline{x} = \lambda \underline{x}$$

$$\underbrace{\lambda_1 = 1, \quad \lambda_2 = 1 \dots \quad \lambda_m = 1}_{\text{Same eigenvalue}}$$

See how we got S before to see the issue

eigenvectors can be linearly independent or linearly dependent $\left\{ \underline{x}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} \quad \underline{x}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} \cdots \quad \underline{x}_n = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} \right.$

1)

- Repeated eigenvalues.
- Linearly dependent eigenvectors.

$A: 2 \times 2$

$$A\underline{x} = \lambda \underline{x}$$

$$\lambda_1 \quad \underline{x}_1$$

$$\lambda_2 = \lambda_1 \quad \underbrace{\underline{x}_1}_{\text{same eigenvector}}$$

$$S = [\underline{x}_1 \quad \underline{x}_1]$$

Suppose its eigenvectors are linearly dependent. S is not invertible and A is non-diagonizable

$$A = T \underbrace{\begin{bmatrix} \lambda_1 & 1 \\ 0 & \lambda_1 \end{bmatrix}}_{} T^{-1}$$

closest we can get

$$\begin{bmatrix} \lambda_1 & 1 \\ 0 & \lambda_1 \end{bmatrix}^2 = \begin{bmatrix} \lambda_1 & 1 \\ 0 & \lambda_1 \end{bmatrix} \begin{bmatrix} \lambda_1 & 1 \\ 0 & \lambda_1 \end{bmatrix}$$

$$= \begin{bmatrix} \lambda_1^2 & 2\lambda_1 \\ 0 & \lambda_1^2 \end{bmatrix}$$

$$\begin{bmatrix} \lambda_1 & 1 \\ 0 & \lambda_1 \end{bmatrix}^3 \qquad \begin{bmatrix} \lambda_1 & 1 \\ 0 & \lambda_1 \end{bmatrix}$$

$$= \begin{bmatrix} \lambda_1^3 & \lambda_1^2 + 2\lambda_1^2 \\ 0 & \lambda_1^3 \end{bmatrix} = \begin{bmatrix} \lambda_1^3 & 3\lambda_1^2 \\ & \lambda_1^3 \end{bmatrix}$$

$$\vdots$$

2)

By grouping the eigenvalues of the same values together we get the Jordan Canonical form.

$$A = T \begin{bmatrix} \lambda_1 & 1 & 0 & & & & \\ 0 & \lambda_1 & 1 & & & & \\ 0 & 0 & \lambda_1 & & & & \\ & & & \lambda_2 & 1 & & \\ & & & 0 & \lambda_2 & & \\ & & & & & \lambda_3 & \\ & & & & & & \lambda_3 \end{bmatrix} T^{-1}$$

Jordan Canonical Form.

$$\mathcal{L}\left[\frac{1}{(s+a)^3}\right]$$

$$3t^2 e^{-at} u(t)$$

II)

Self-adjoint. $A^* = A$ $\quad A^T = A$

$A^H = A$

D)

lemma: The eigenvalues of a self adjoint matrix are real.

Pf: $A\underline{x} = \lambda \underline{x}$

$$\langle A\underline{x}, \underline{x}\rangle = \langle \lambda \underline{x}, \underline{x}\rangle = \lambda \underbrace{\langle \underline{x}, \underline{x}\rangle}_{\text{real}}$$ – this is always real

$$\|$$

$$\langle \underline{x}, A^* \underline{x}\rangle = \langle \underline{x}, A\underline{x}\rangle = \langle \underline{x}, \lambda \underline{x}\rangle = \lambda^* \langle \underline{x}, \underline{x}\rangle$$

$$\lambda \langle x, x\rangle = \lambda^* \langle \underline{x}, \underline{x}\rangle$$

$$\lambda = \lambda^* \longleftarrow \lambda \text{ must be real.}$$

B)

lemma: If $A = A^*$ The eigenvectors corresponding to distinct eigenvalues are orthogonal.

Pf: $A\underline{x}_1 = \lambda_1 \underline{x}_1, \quad A\underline{x}_2 = \lambda_2 \underline{x}_2, \quad$ and $\lambda_1 \neq \lambda_2$

$$\langle A\underline{x}_1, \underline{x}_2\rangle = \langle \lambda_1 \underline{x}_1, \underline{x}_2\rangle = \lambda_1 \langle \underline{x}_1, \underline{x}_2\rangle$$

$$\|$$

$$\langle \underline{x}_1, A^* \underline{x}_2\rangle = \langle \underline{x}_1, A\underline{x}_2\rangle = \langle \underline{x}_1, \lambda_2 \underline{x}_2\rangle$$

$$= \lambda_2 \langle \underline{x}_1, \underline{x}_2\rangle$$

Nasu Caution 1 Dana 4

$$\lambda_1 \langle \underline{x}_1, \underline{x}_2 \rangle = \lambda_2 \langle \underline{x}_1, \underline{x}_2 \rangle$$

$$\underbrace{(\lambda_1 - \lambda_2)}_{\neq 0} \langle \underline{x}_1, \underline{x}_2 \rangle = 0$$

so $\langle \underline{x}_1, \underline{x}_2 \rangle = 0$.

C) Stack all the eigenvalues and eigenvectors

Start as before:

$$A \underbrace{[\underline{x}_1 \ \underline{x}_2 \ \cdots \ \underline{x}_m]}_{U} = \underbrace{[\underline{x}_1 \ \underline{x}_2 \ \cdots \ \underline{x}_m]}_{U} \underbrace{\begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_m \end{bmatrix}}_{\Lambda}$$

Assume $\underline{x}_i$ are normalized, so $\|\underline{x}_i\| = 1$

$$AU = U\Lambda$$

Notice that $U^H U = I$ $\quad$ U is unitary

$$\boxed{A = U \Lambda U^H} \qquad \boxed{A = U \Lambda U^T}$$

D) Thm: Every $m \times m$ self adjoint matrix can be written as

$$A = U \Lambda U^H$$

where Λ is a diagonal matrix.

1) Write out the theorem

$$A = \begin{bmatrix} | & | & & | \\ \underline{u}_1 & \underline{u}_2 & \cdots & \underline{u}_m \\ | & | & & | \end{bmatrix} \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_m \end{bmatrix} \begin{bmatrix} - & u_1^H & - \\ - & u_2^H & - \\ & \vdots & \\ - & u_m^H & - \end{bmatrix}$$

$$= \lambda_1 \underline{u}_1 \underline{u}_1^H + \lambda_2 \underline{u}_2 \underline{u}_2^H + \cdots + \lambda_m \underline{u}_m \underline{u}_m^H$$

$$\boxed{A = \sum_{i=1}^{m} \lambda_i \underline{u}_i \underline{u}_i^H}$$

$\begin{bmatrix} | \\ | \\ | \end{bmatrix} [\ \text{———} \ ]$

$$A = U \Lambda U^H$$

$$A^{-1} = (U \Lambda U^H)^{-1}$$

$$= U^{-H} \Lambda^{-1} U^{-1}$$

$$= U \Lambda^{-1} U^H$$

$$U^H U = I$$

$$U^{-1} = U^H$$

$$U = U^{-H}$$